

Phishing Email Detection using Python

Python Program:

```
import re

def check_email(sender, subject, body):
    flags = []

    if re.search(r"(urgent|verify your account|suspended|click here)",
body, re.IGNORECASE):
        flags.append("Urgency/pressure language detected")

    if re.search(r"\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}", body):
        flags.append("Raw IP address link found in body")

    domain = sender.split("@")[-1]

    if any(k in domain.lower() for k in ["secure", "verify", "update"])
and "@gmail" not in sender:
        flags.append("Suspicious sender domain naming pattern")

    return flags

emails = [
("support@paypal.com", "Your monthly statement", "Please find your
statement attached."),
("alert@paypal-secure-verify.com", "URGENT: Verify your account", "Click
here: http://192.168.10.5/login")
]

for sender, subject, body in emails:
    issues = check_email(sender, subject, body)
    verdict = "PHISHING SUSPECTED" if issues else "Looks legitimate"

    print("From:", sender)
    print("Subject:", subject)
    print("Verdict:", verdict)
    for issue in issues:
        print("-", issue)
```

Sample Output:

```
From: support@paypal.com
Subject: Your monthly statement
Verdict: Looks legitimate
```

```
From: alert@paypal-secure-verify.com
Subject: URGENT: Verify your account
Verdict: PHISHING SUSPECTED
- Urgency/pressure language detected
- Raw IP address link found in body
- Suspicious sender domain naming pattern
```

Explanation:

This program demonstrates a simple phishing email detector using Python and regular expressions. It analyzes email content for common phishing indicators such as urgency phrases, links containing raw IP addresses, and suspicious sender domain names. If one or more indicators are detected, the email is marked as **PHISHING SUSPECTED**; otherwise, it is considered likely legitimate. This

illustrates basic rule-based phishing detection used in cybersecurity education.